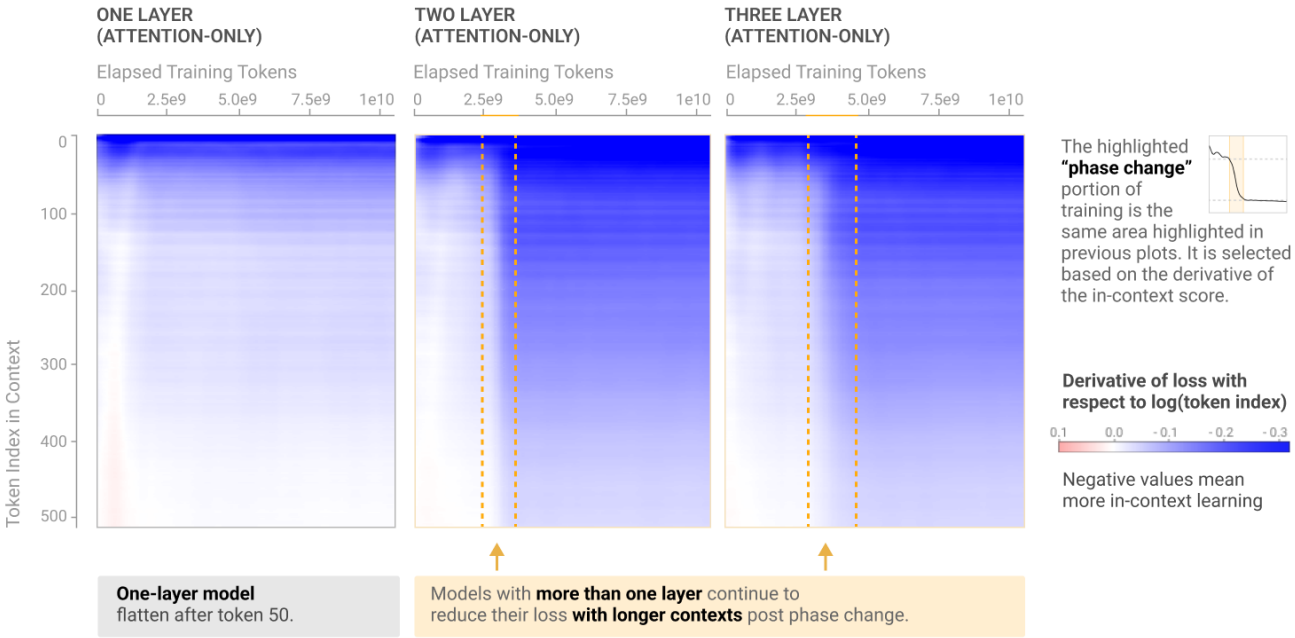
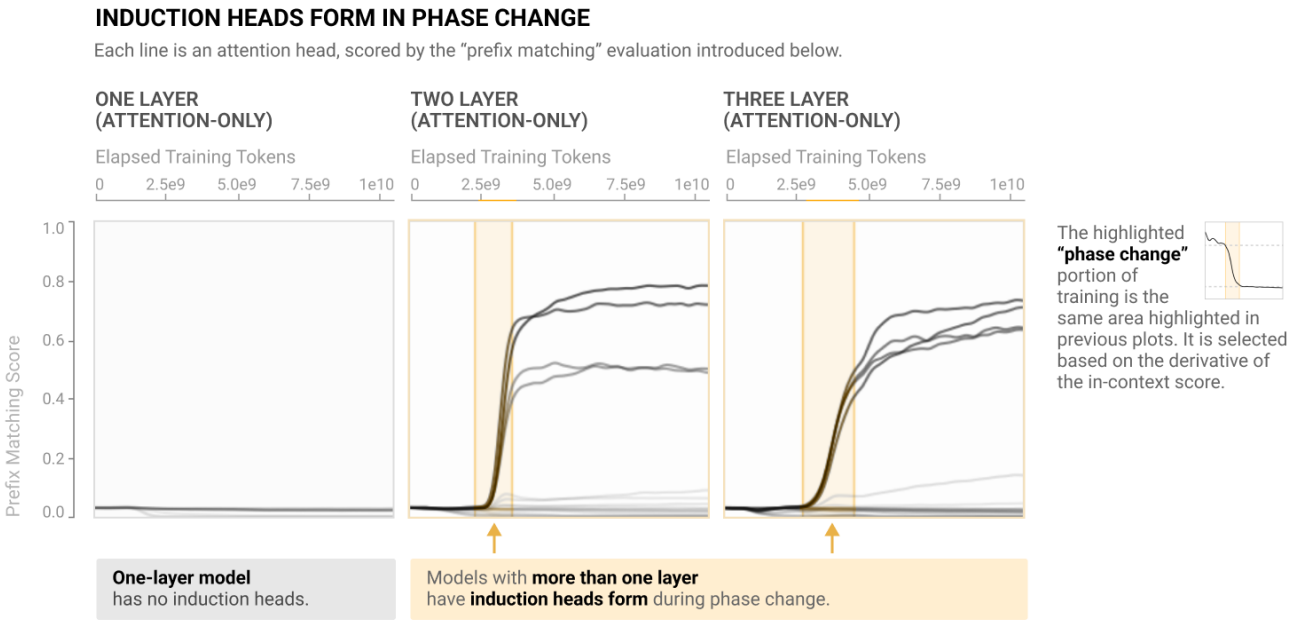
One might wonder if the sudden increase is somehow an artifact of the choice to define in-context learning in terms of the difference between the 500th and 50th tokens. We'll discuss this in more depth later. But for now, an easy way to see that this is a robust phenomena is to look at the derivative of loss with respect to the logarithm token index in context. You can think of this as measuring something like "in-context learning per ε% increase in context length." We can visualize this on a 2D plot, where one axis is the amount of training that has elapsed, the other is the token index being predicted. Before the phase change, loss largely stops improving around token 50, but after the phase change, loss continues to improve past that point.

DERIVATIVE OF LOSS WITH RESPECT TO LOG TOKEN INDEX

The rate at which loss decreases with increasing token index can be thought of as something like "in-context learning per token". This appears to be most naturally measured with respect to the log number of tokens.



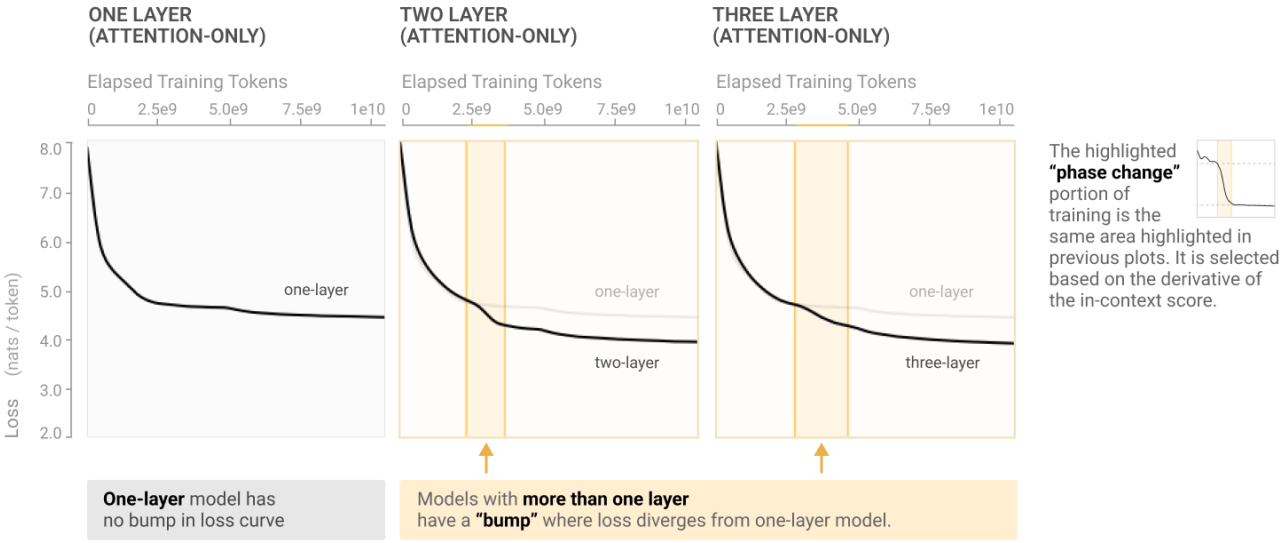
It turns out that a sudden improvement in in-context learning isn't the only thing that changes in this window. If we go through the attention heads of a model and and score them for whether they are induction heads (using a *prefix matching score* which measures their ability to perform the task we used to define induction heads in Key Concepts), we find that induction heads form abruptly during exactly the same window where in-context learning develops (figure below). Again we show only a few models here, but a full set is shown in the Model Analysis Table. The exception is the one-layer model, where induction heads never form – just as in-context learning never substantially develops for the one-layer model.



This already strongly suggests some connection between induction heads and in-context learning, but beyond just that, it appears this window is a pivotal point for the training process in general: whatever's occurring is visible as a bump on the training curve (figure below). It is in fact the only place in training where the loss is not convex (monotonically decreasing in slope).

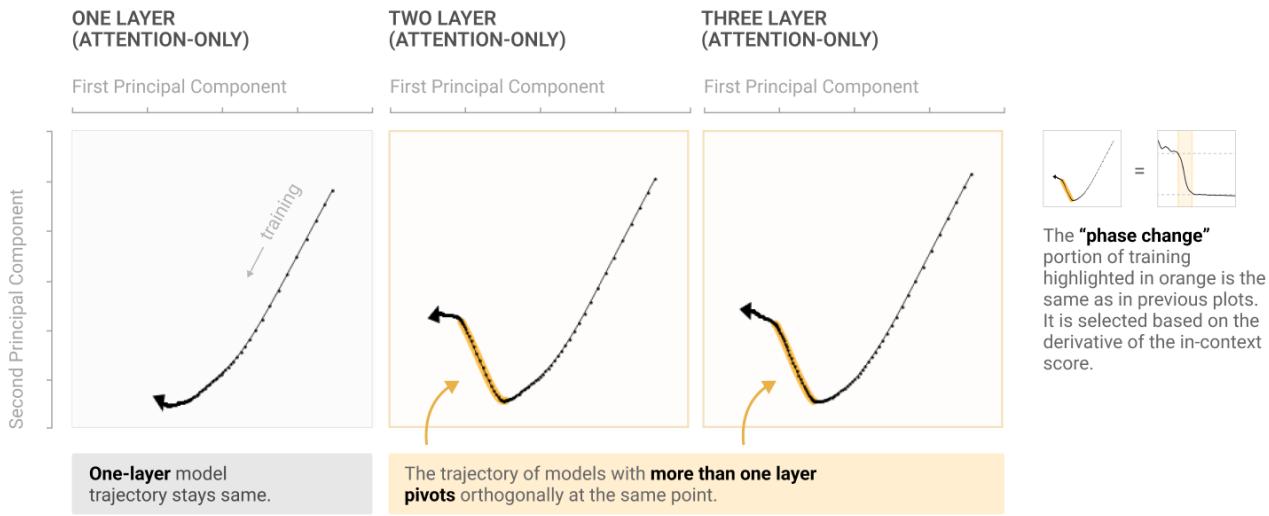
That might not sound significant, but the loss curve is averaging over many thousands of tokens. Many behaviors people find interesting in language models, such as the emergence of arithmetic, would be microscopic on the loss curve. For something to be visible at that scale suggests it's a widespread, major change in model behavior. This shift also appears to be the first point where, at least for small models, the loss curve diverges from a one-layer model – which does not display the bump, just as it does not display the other abrupt changes.

LOSS CURVES DIVERGE DURING PHASE CHANGE



We can also apply principal components analysis (PCA) to the per-token losses, as described in per-token-loss analysis, which allows us to summarize the main dimensions of variation in how several models' predictions vary over the course of training.

Below we show the first two principal components of these models' predictions, with the golden outline highlighting the same interval shown above, when in-context learning abruptly improved. We see that the training trajectories pivot during exactly the same window where the other changes happen. In some sense, whatever is occurring when in-context learning improves is the primary deviation from the basic trajectory our transformers follow during the course of their training. Once again the only exception is the one-layer model – where induction heads cannot form and in-context learning does not improve.

PER-TOKEN LOSS PRINCIPAL COMPONENT ANALYSIS

This method is described above, but roughly shows models' training trajectories on the two dimensions there is the most behavioral variance. Note how it abruptly pivots for models with more than one layer.